

Explainable Federated Learning Framework for Privacy-Aware Breast Cancer Diagnosis

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Abstract—Breast cancer is one of the most common cancers among women worldwide, making accurate and timely diagnosis crucial for improving clinical outcomes. Although machine learning models have demonstrated strong diagnostic performance, centralized learning requires the collection of sensitive patient data, raising significant privacy concerns. This study proposes an explainable federated learning framework for privacy-aware breast cancer diagnosis using the Breast Cancer Wisconsin (Diagnostic) (WDBC) dataset. The proposed framework evaluates Logistic Regression, Random Forest, and XGBoost under centralized and federated learning settings across four simulated healthcare clients. A Federated Soft Voting Ensemble is further employed to combine the complementary strengths of the individual classifiers. Among the individual federated models, Logistic Regression achieved the highest accuracy of 98.25%, while the Federated Soft Voting Ensemble provided a balanced performance with 97.37% accuracy, 100.00% precision, 92.86% recall, 96.30% F1-score, and a ROC-AUC of 0.998. To improve model interpretability, SHAP and LIME are integrated to provide complementary global feature attributions and local instance-level explanations. The results demonstrate that the proposed framework achieves predictive performance comparable to centralized learning while reducing the need to centralize sensitive patient data. Overall, the proposed framework demonstrates the potential of integrating lightweight machine learning, Federated Learning, and Explainable AI to develop accurate, privacy-preserving, and interpretable breast cancer diagnostic systems.

Index Terms—Breast Cancer Diagnosis, Federated Learning, Explainable AI (XAI), Machine Learning, Ensemble Learning, SHAP, LIME, Privacy-Aware Healthcare

I. INTRODUCTION

Breast cancer is the most common cancer among women worldwide, with approximately 2.3 million new cases reported in 2022 [12]. It remains one of the leading causes of cancer-related mortality among women, posing a significant global public health challenge. As illustrated in Fig. 1, breast cancer has the highest age-standardized incidence rate among cancers in India, highlighting its growing burden on the healthcare system. According to GLOBOCAN 2022 estimates, breast cancer accounted for more than 98,000 deaths in India, highlighting the substantial burden of the disease. These statistics underscore the importance of accurate and timely diagnosis to support improved clinical outcomes.

A. Current Diagnosis Methods

To facilitate early breast cancer diagnosis, clinicians rely on medical imaging techniques such as mammography, ultrasonography, and magnetic resonance imaging (MRI). Although these imaging modalities play a crucial role in disease

Age-Standardized Rate (World) per 100 000, Incidence, Both sexes, in 2022
India
(Top 15 cancer sites)

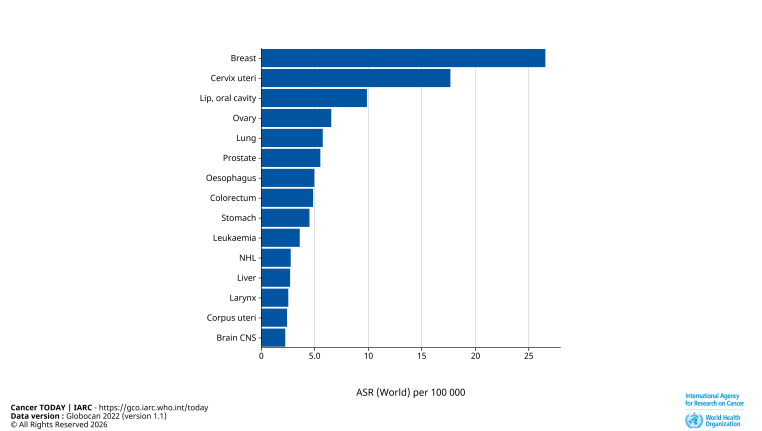


Fig. 1. Age-standardized incidence rates of the fifteen most common cancers in India (GLOBOCAN 2022). Source: GLOBOCAN 2022.

detection, their accuracy often depends on the expertise and experience of clinicians, creating a need for computer-aided diagnostic support systems that assist clinicians in making consistent diagnostic decisions.

B. AI and Machine Learning

Artificial Intelligence (AI) and Machine Learning (ML) have significantly advanced computer-aided diagnosis by learning complex patterns from medical data. In breast cancer diagnosis, ML models can assist clinicians by improving prediction accuracy, reducing diagnostic variability, and supporting early disease detection [10], [11]. Despite these advantages, conventional machine learning approaches rely on centralized data collection, making them difficult to deploy in privacy-sensitive healthcare settings.

C. Need for Federated Learning

Despite the success of classical machine learning algorithms in predictive tasks, they rely on centralized training, where patient data from different healthcare institutions are collected and stored on a central server. Since healthcare data contain sensitive patient information, this approach raises privacy, security, and regulatory concerns, including compliance with HIPAA and GDPR. These challenges limit collaboration among healthcare institutions for developing reliable diagnostic models.

To address these challenges, Federated Learning (FL) enables hospitals and clinical systems to train models locally while sharing only model updates with a central server [1]. This allows multiple healthcare institutions to collaboratively develop a global model without centralizing sensitive patient data. However, despite its privacy-aware design, federated learning models often lack interpretability, making their predictions difficult for clinicians to understand.

D. Need for Explainability

Although Federated Learning offers significant advantages by avoiding centralized storage of patient data, limited model explainability remains a major barrier to its adoption in clinical practice [2]. While accurate predictions are important, clinicians must also understand the factors influencing a model's predictions before adopting AI-based systems in clinical practice. Explainable Artificial Intelligence (XAI) techniques, such as SHAP and LIME, enhance model transparency by providing global and local explanations of model predictions [7]. However, most existing studies primarily emphasize predictive performance while giving limited attention to explainability within federated learning environments. These limitations highlight the need for a framework that simultaneously supports predictive performance, data privacy, and model interpretability [5].

E. Proposed Framework

Based on the research gaps identified in the literature, this study proposes an explainable federated learning framework for privacy-aware breast cancer diagnosis, as illustrated in Fig. 2. The framework integrates Federated Learning with Explainable Artificial Intelligence (XAI) to enable distributed model training while preserving data locality and improving model transparency. It compares multiple machine learning algorithms under centralized and federated learning settings, while SHAP and LIME provide complementary global and local explanations for model predictions.

Unlike existing studies that primarily focus on predictive performance or a single explainability technique, the proposed framework combines Federated Learning with both SHAP and LIME while evaluating multiple machine learning models [3]-[9]. This lightweight framework aims to support privacy-aware, transparent, and interpretable AI-assisted breast cancer diagnosis.

F. Contributions

The main contributions of this paper are:

- Development of an explainable federated learning framework for privacy-aware breast cancer diagnosis.
- Comparative evaluation of Logistic Regression, Random Forest, and XGBoost under both centralized and federated learning settings.
- Integration of SHAP and LIME to provide complementary global and local explanations, enhancing model transparency and interpretability.
- Evaluation of the proposed framework, including global and

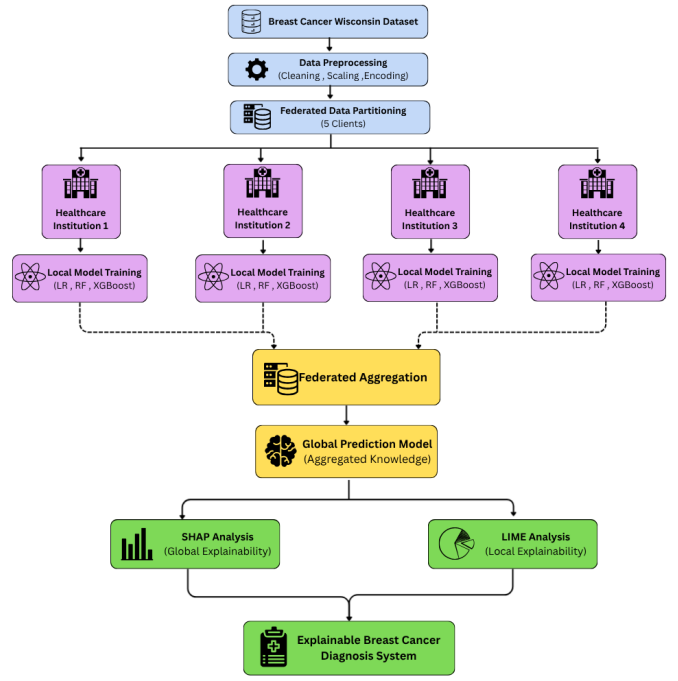


Fig. 2. Conceptual architecture of the proposed privacy-aware explainable federated learning framework.

local explainability analysis.

G. Paper Organization

The remainder of this paper is organized as follows. Section II reviews the related literature. Section III presents the proposed methodology. Section IV presents the experimental results and discussion. Finally, Section V concludes the paper and discusses future research directions.

II. LITERATURE REVIEW

The development of privacy-aware diagnostic frameworks requires a comprehensive understanding of Federated Learning (FL) and Explainable Artificial Intelligence (XAI). This section reviews recent research in these domains by analyzing existing methodologies, highlighting their key contributions, and identifying their limitations. Based on this analysis, the section summarizes the common findings in the literature and outlines the research gaps that motivate the proposed explainable federated learning framework.

A. Federated Learning (FL) in Healthcare

Federated Learning (FL) has emerged as a promising approach for addressing data privacy and governance challenges in healthcare by enabling collaborative model development without requiring institutions to centralize sensitive patient data [1].

- **Federated Learning in Healthcare:** The survey presented in [1] provides a comprehensive overview of Federated Learning applications in healthcare, emphasizing

its ability to facilitate decentralized model training while retaining raw patient data within local institutions. The study also discusses practical challenges associated with federated deployments, including heterogeneous client data, communication overhead, and model convergence.

- **Federated Breast Cancer Classification:** The study in [9] investigates the integration of Federated Learning and Explainable Artificial Intelligence (XAI) for breast cancer classification. The authors demonstrate the feasibility of combining collaborative model training with explainability, highlighting the importance of interpretable machine learning models for healthcare applications.

Common Findings: The reviewed studies indicate that Federated Learning supports collaborative model development while reducing the need to centralize patient data [1]. Recent research has also explored the integration of Federated Learning with explainability techniques for breast cancer classification [9]. However, challenges related to heterogeneous client data, communication overhead, and model convergence continue to limit practical deployment. In addition, relatively few studies have investigated classical machine-learning frameworks that jointly integrate federated learning and explainability. These observations motivate the development of the proposed privacy-aware and explainable federated learning framework.

B. Explainable AI (XAI) in Clinical Pipelines

As machine learning models become increasingly complex, understanding the factors influencing their predictions has become essential for improving model transparency and supporting AI-assisted clinical decision support. Explainable Artificial Intelligence (XAI) techniques address this challenge by providing interpretable explanations that help users understand how model predictions are generated [2].

- **Explainability in Healthcare:** The survey presented in [2] reviews a range of XAI techniques applied in healthcare and emphasizes the importance of model interpretability for improving the transparency of AI-assisted decision support. Among the widely adopted post-hoc explainability methods, SHAP and LIME provide complementary approaches for interpreting machine learning model predictions [7].
- **Explainability in Breast Cancer Diagnosis:** Recent studies have investigated the application of XAI techniques to breast cancer detection and diagnosis [10], [11]. These studies demonstrate how explainability can identify influential diagnostic features and improve the transparency of machine learning-based breast cancer prediction systems.

Common Findings: Collectively, these studies demonstrate that XAI techniques improve the transparency and interpretability of machine learning models by providing insights into the features influencing model predictions [2], [7]. Recent research on breast cancer diagnosis has further demonstrated the value of explainability for interpreting machine learning predictions [10], [11]. However, most studies investigate ex-

plainability independently of decentralized learning environments, leaving opportunities to integrate complementary XAI techniques within federated healthcare frameworks. Motivated by these observations, the proposed framework incorporates both SHAP and LIME within a Federated Learning framework to provide complementary global and local explanations while supporting privacy-aware collaborative learning.

C. Explainable Federated Learning Frameworks

Recent research has focused on integrating Federated Learning with Explainable Artificial Intelligence (XAI), leading to the development of Explainable Federated Learning (XFL) frameworks that aim to address both data privacy and model interpretability [5], [8].

- **Explainable Federated Learning in Healthcare:** Existing studies have explored the integration of Federated Learning and explainability across various healthcare applications, including liver disease prediction [3], heart disease diagnosis [4], and other privacy-sensitive healthcare settings [5], [6]. These studies demonstrate the growing importance of combining decentralized model training with explainability to improve the transparency of AI-assisted healthcare applications.
- **Explainable Federated Learning for Breast Cancer:** Recent research has also investigated explainable federated learning for breast cancer classification [9]. These studies demonstrate the feasibility of combining collaborative model training with explainability for breast cancer diagnosis. However, opportunities remain to explore lightweight machine learning models and complementary explainability techniques within a unified federated framework.

Common Findings: The reviewed studies demonstrate that integrating explainability techniques with Federated Learning can improve model transparency while reducing the need to centralize sensitive healthcare data [3]–[6], [8]. However, limited attention has been given to classical machine learning frameworks that combine multiple classifiers with complementary global and local explainability techniques for structured breast cancer data. These observations motivate the proposed framework, which integrates classical machine learning models with SHAP and LIME in a federated setting to support privacy-aware and interpretable breast cancer diagnosis.

D. Research Gap

The literature review identifies several research gaps that motivate the development of the proposed framework:

- **Limited Integration of Explainability:** Many federated healthcare studies focus on predictive performance and privacy-aware distributed learning, with comparatively less emphasis on model transparency and post-hoc explainability.
- **Limited Comparative Evaluation of XAI Techniques:** To the best of our knowledge, limited studies provide a comparative analysis of complementary explainability

methods, such as SHAP and LIME, within federated healthcare environments.

- **Limited Exploration of Lightweight Machine Learning Architectures:** Existing research frequently rely on complex deep learning models, while the integration of lightweight classical machine learning models within explainable federated frameworks remains relatively underexplored.
- **Limited Comparative Analysis of Centralized and Federated Learning:** Comparatively few studies have systematically evaluated centralized and federated learning using multiple machine learning models on structured breast cancer datasets.

To address these identified research gaps, the proposed framework integrates classical machine learning models, Federated Learning, and complementary explainability techniques to investigate privacy-aware and interpretable breast cancer diagnosis in a simulated federated environment.

E. Research Contributions

To address the identified research gaps, this work proposes a lightweight and explainable diagnostic framework evaluated on the Breast Cancer Wisconsin (Diagnostic) (WDBC) dataset, a widely used benchmark dataset for breast cancer diagnosis.

- **Comparative Evaluation of Centralized and Federated Learning:** The proposed framework systematically evaluates and compares centralized and federated learning using three classical machine learning models: Logistic Regression, Random Forest (RF), and XGBoost.
- **Lightweight Ensemble Architecture:** A lightweight Federated Soft Voting Ensemble is developed by combining the complementary strengths of linear and tree-based classifiers across four simulated client silos.
- **Dual-Layer Explainability Framework (SHAP + LIME):** To improve model interpretability in the federated setting, the proposed framework integrates complementary explainability techniques. SHAP is used to provide global feature attribution, while LIME generates local explanations for individual model predictions.
- **Framework Justification:** The proposed framework combines classical machine learning models, Federated Learning, and complementary explainability techniques to address the identified research gaps while supporting interpretable AI-assisted diagnostic decision support.

Table I summarizes the representative studies discussed in this section and highlights their key contributions, identified limitations, and the research gaps addressed by the proposed framework.

III. METHODOLOGY

A. Overview of Proposed Framework

The overall workflow of the proposed explainable federated learning framework is illustrated in Fig. 3. The Breast Cancer Wisconsin (Diagnostic) (WDBC) dataset is first preprocessed

and divided into training and testing sets using an 80:20 train-test split. The training data are then partitioned among four simulated healthcare clients to emulate a federated learning environment, while the global test set is reserved for evaluating the global models. Each client independently trains Logistic Regression, Random Forest, and XGBoost models without exchanging raw patient data, reducing the need for centralized data storage while supporting privacy-aware distributed model training. The locally trained Logistic Regression models are aggregated using one-shot FedAvg-style parameter averaging. In contrast, Random Forest and XGBoost models are not aggregated at the parameter level. Instead, their class probabilities are averaged across the four clients at the prediction level, representing a one-shot federated ensemble setting rather than conventional parameter-aggregated federated learning [13]. The resulting global models are evaluated on the global test set, and SHAP and LIME are used to generate global and local explanations, improving model transparency and supporting interpretation [7].

B. Dataset Description

The main characteristics of the WDBC dataset are summarized in Table II.

TABLE II
CHARACTERISTICS OF THE BREAST CANCER WISCONSIN (DIAGNOSTIC) (WDBC) DATASET.

Attribute	Description
Dataset	Breast Cancer Wisconsin (Diagnostic) (WDBC)
Source	UCI Machine Learning Repository
Total Samples	569
Features	30 numerical features
Target Classes	Benign (357), Malignant (212)
Missing Values	0

C. Data Preprocessing

Before federated training, the Breast Cancer Wisconsin (Diagnostic) (WDBC) dataset underwent preprocessing to ensure consistent model training. The identifier (ID) attribute was removed because it does not contain information relevant to disease prediction, while the diagnosis labels were encoded into binary classes (malignant = 1 and benign = 0). The dataset was then divided into training and testing subsets using an 80:20 stratified train-test split to preserve the original class distribution. Feature standardization was performed using StandardScaler, which was fitted only on the training data and subsequently applied to the global test set to prevent data leakage. As part of the simulation setup, the standardized training data were partitioned among four federated clients for local model training. This global preprocessing step was adopted to ensure consistent feature scaling across clients and is considered a simulation assumption rather than a deployment-ready federated preprocessing strategy. In a real-world deployment, the scaling statistics would need to be obtained through federated aggregation of training statistics or through client-local preprocessing to avoid centralized access to the complete training data.

TABLE I
SUMMARY OF REPRESENTATIVE STUDIES ON FEDERATED LEARNING AND EXPLAINABLE ARTIFICIAL INTELLIGENCE IN HEALTHCARE.

Ref.	Research Focus	Key Contribution	Limitation / Research Gap
[1]	FL in Healthcare Survey (2026)	Comprehensive review of Federated Learning paradigms, challenges, and healthcare applications.	Lacks practical implementation and real-world validation.
[2]	XAI in Healthcare Survey (2023)	Categorized major XAI techniques including SHAP and LIME for clinical AI.	No standardized evaluation framework for trustworthy explanations.
[3]	FL-XAI for Liver Disease (2025)	Combined Federated Learning with SHAP for privacy-preserving diagnosis and achieved high predictive performance.	Evaluated on a single disease with simulated clients.
[9]	Federated XAI for Breast Cancer (2024)	Applied Federated Learning with SHAP on breast cancer datasets while preserving privacy.	Limited model diversity and simplified federated architecture.
[5]	Secure Explainable FL (2024)	Integrated security mechanisms with explainable federated learning.	Increased computational and communication overhead.
[6]	FL-XAI for Parkinson's Disease (2024)	Demonstrated privacy-preserving prediction with federated SHAP explanations.	Performance degraded under highly non-IID client distributions.
[10]	Explainable AI in Breast Cancer Review (2024)	Reviewed SHAP, LIME, Grad-CAM, and other XAI methods for breast cancer diagnosis.	Limited investigation of XAI in decentralized healthcare environments.
[8]	Explainable Federated Learning Survey (2026)	Summarized existing FL-XAI techniques and future research directions.	Few real-world explainable federated healthcare deployments.

D. Federated Data Partitioning

TABLE III
CLASS DISTRIBUTION OF TRAINING SAMPLES ACROSS THE SIMULATED FEDERATED CLIENTS.

Client	Benign	Malignant	Total
Client 1	73	41	114
Client 2	73	41	114
Client 3	67	47	114
Client 4	72	41	113
Total	285	170	455

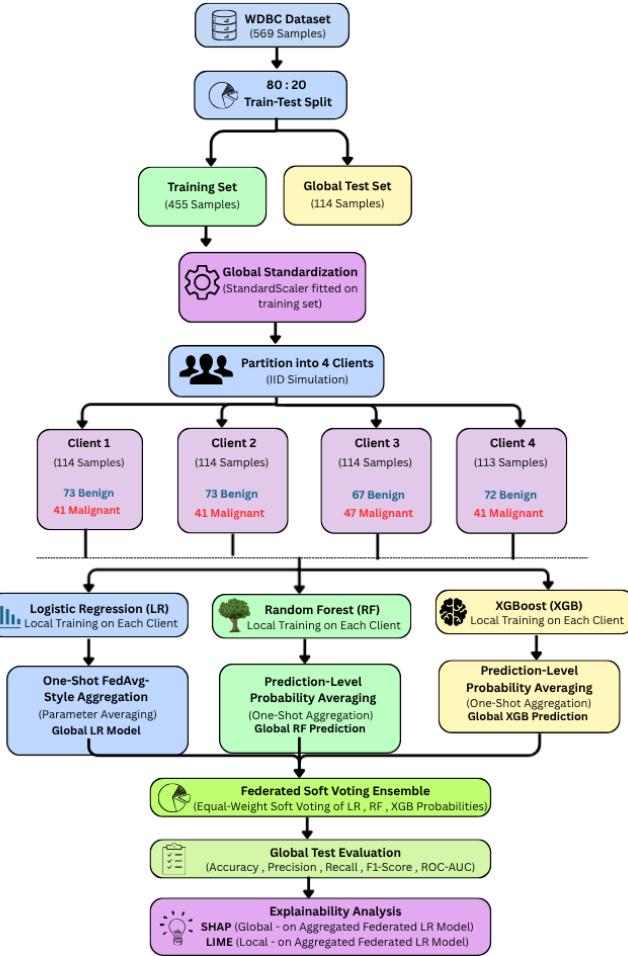


Fig. 3. Workflow of the proposed privacy-aware explainable federated learning framework illustrating preprocessing, client partitioning, model-specific aggregation, ensemble learning, evaluation, and explainability.

To simulate a federated healthcare environment, the pre-processed training dataset was partitioned into four mutually exclusive subsets, each representing an independent healthcare institution. As described in the preprocessing stage, partitioning was performed after global feature standardization to maintain consistent feature scaling across clients. This preprocessing strategy represents a simulation assumption rather than a deployment-ready federated preprocessing strategy. Each client independently trained its local models using only its assigned subset without access to data from other clients. The global test set was reserved exclusively for evaluating the aggregated global model predictions. Table 3 summarizes the benign and malignant class distribution across the four simulated clients. Since the client subsets were generated by randomly shuffling the stratified training set before partitioning, the resulting distribution approximates an IID federated simulation. The 80:20 stratified train-test split used random state 42. For client partitioning, the standardized training indices were shuffled using NumPy random seed 42 and divided into four approximately equal client subsets using array-based partitioning.

E. Federated Learning Architecture

The proposed federated learning framework consists of four simulated healthcare institutions and a central aggregation

server. Following data partitioning, each client independently trained Logistic Regression, Random Forest, and XGBoost models using only its assigned local dataset, without exchanging raw patient data. The locally trained Logistic Regression models were aggregated at the central server using one-shot FedAvg-style parameter averaging. Since all four clients contained nearly equal numbers of training samples (113–114 samples per client), simple averaging of the locally learned coefficients and intercepts was used to obtain the global Logistic Regression model. In contrast, Random Forest and XGBoost models were not aggregated at the parameter level. Instead, each locally trained model generated class probabilities for the global test set, and the corresponding probabilities were averaged across the four clients to obtain aggregated Random Forest and XGBoost predictions. This represents a one-shot prediction-level aggregation strategy rather than conventional parameter-aggregated federated learning. In this simulated setup, the global test samples were evaluated by the locally trained models, and only the resulting class probabilities were considered for aggregation; raw client training data were not exchanged. This proposed framework reduces the need to centralize raw patient data and supports privacy-aware distributed model training. However, this privacy-aware design does not provide a formal privacy guarantee, as the current implementation does not incorporate mechanisms such as differential privacy, secure aggregation, or encryption. In this study, all four clients participated in a single communication round without iterative client-server optimization. The resulting global models were subsequently evaluated using the global test set.

F. Machine Learning Models and Ensemble Learning

Three supervised machine learning algorithms, namely Logistic Regression (LR), Random Forest (RF), and Extreme Gradient Boosting (XGBoost), were selected as the base models for the proposed framework. These models were chosen because they represent complementary learning paradigms, representing linear, bagging, and boosting paradigms, thereby enabling a comparative evaluation of different classification strategies for breast cancer diagnosis. Each model was trained independently on the local dataset of every simulated federated client using identical training settings to ensure a fair and consistent performance comparison. Logistic Regression models were aggregated using one-shot FedAvg-style parameter averaging, whereas Random Forest and XGBoost used one-shot prediction-level probability averaging across the four clients, as described in Section 3.5. The implementation parameters for each classifier are summarized in Table 4. To further investigate the effect of combining complementary classifiers, a Soft Voting Ensemble was constructed by combining the class probabilities generated by Logistic Regression, Random Forest, and XGBoost. Unlike hard voting, which selects the majority class label, soft voting computes the average of the predicted class probabilities before assigning the final class label. In this study, equal weights were assigned to all three classifiers, and no additional probability calibration was

TABLE IV
HYPERPARAMETER SETTINGS OF THE MACHINE LEARNING MODELS.

Model	Hyperparameter Settings
Logistic Regression	max_iter = 1000, solver = lbfgs (default), C = 1.0 (default), random_state = 42
Random Forest	n_estimators = 50, max_depth = None (default), class_weight = None (default), random_state = 42
XGBoost	n_estimators = 50, learning_rate = 0.3 (default), max_depth = 6 (default), subsample = 1.0 (default), eval_metric = logloss, random_state = 42

applied. This approach leverages the complementary strengths of the individual classifiers, providing a combined prediction strategy for both the centralized and simulated federated evaluation settings.

G. Model Evaluation and Explainability

The proposed framework was evaluated using five widely adopted classification metrics: Accuracy, Precision, Recall, F1-score, and Receiver Operating Characteristic–Area Under the Curve (ROC-AUC). For binary classification, the default decision threshold of 0.5 was used for Logistic Regression predictions, while Random Forest, XGBoost, and the soft-voting ensemble assigned the class with the highest predicted probability. Together, these metrics assess overall predictive performance, the reliability of positive predictions, the ability to correctly identify malignant cases, and the balance between precision and recall. A brief description of each evaluation metric is presented in Table 5.

TABLE V
PERFORMANCE EVALUATION METRICS USED IN THIS STUDY.

Metric	Purpose
Accuracy	Measures overall classification accuracy
Precision	Evaluates correctness of malignant predictions
Recall	Measures detection capability of malignant cases
F1-Score	Balances precision and recall
ROC-AUC	Measures discriminative performance

To improve the transparency and interpretability of the proposed framework, Explainable Artificial Intelligence (XAI) techniques were incorporated [2]. SHAP (SHapley Additive exPlanations) was applied to the aggregated federated Logistic Regression model to quantify feature contributions and provide global feature importance through SHAP summary analysis. LIME (Local Interpretable Model-agnostic Explanations) was applied to the same aggregated federated Logistic Regression

model to generate instance-level explanations for individual predictions [7]. Together, these techniques provide global and local insights into model behavior, supporting the interpretation of AI-assisted diagnostic decisions. Since the framework was evaluated using a public benchmark dataset with simulated federated clients, these explanations are intended to demonstrate the decision-support potential of the proposed framework rather than imply direct clinical deployment. Table 6 summarizes the explainability methods used in this study.

TABLE VI
EXPLAINABILITY TECHNIQUES INTEGRATED INTO THE PROPOSED FRAMEWORK.

Technique	Scope	Explanation
SHAP	Global Feature Attribution	Quantifies feature contributions
LIME	Local	Explains individual predictions

IV. RESULTS AND DISCUSSION

A. Performance Evaluation in a Centralized Setting

The centralized performance of Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB) was evaluated using an 80:20 train-test split. The experimental results, including Accuracy, Precision, Recall, F1-score, and ROC-AUC, are presented in Table 7.

TABLE VII
PERFORMANCE OF CENTRALIZED MACHINE LEARNING MODELS.

Model	Acc.	Prec.	Rec.	F1	AUC
LR	96.49	97.50	92.86	95.12	0.996
RF	97.37	100	92.86	96.30	0.993
XGB	97.37	100	92.86	96.30	0.994

As shown in Table 7, Random Forest and XGBoost achieved the highest classification accuracy (97.37%), outperforming Logistic Regression (96.49%). Both ensemble models also obtained higher F1-scores (96.30%), indicating a better balance between precision and recall. Logistic Regression achieved the highest ROC-AUC (0.996), demonstrating excellent discriminative ability despite its slightly lower classification accuracy. These results indicate that all three models achieved high predictive performance in the centralized setting and provide a reference for comparison with the proposed simulated federated learning framework.

B. Cross-Validation Performance Analysis

To further assess the generalization performance of the proposed models, stratified five-fold cross-validation was performed on the centralized dataset. The mean Accuracy, Precision, Recall, F1-score, and ROC-AUC values across the five folds are presented in Table 8.

As shown in Table 8, all three models achieved consistently high performance with relatively small standard deviations across the evaluation metrics, indicating stable performance

across different training and validation splits. Logistic Regression achieved the highest mean Accuracy ($97.37 \pm 1.66\%$) and ROC-AUC (0.995 ± 0.005), while Random Forest and XGBoost also demonstrated competitive and consistent performance. These results indicate consistent performance across the five cross-validation folds within the benchmark dataset, with relatively small variation across most evaluation metrics.

C. Performance Evaluation of Federated Models

The proposed federated learning framework was evaluated using four simulated healthcare clients. The experimental results for Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB) are presented in Table 9. Overall, the federated models achieved performance comparable to their centralized counterparts despite being trained on distributed client datasets. Among the three models, Logistic Regression achieved the highest federated accuracy (98.25%) and ROC-AUC (0.998), while Random Forest and XGBoost also demonstrated strong predictive performance. The higher accuracy of the federated Logistic Regression model compared with its centralized counterpart may be attributed to the averaging of independently trained client models, which can reduce the influence of client-specific variations when the client datasets are nearly balanced in size and class distribution. However, this observation is specific to this experimental setup and was not evaluated using repeated trials or statistical significance testing. Overall, these results indicate that the proposed framework can achieve competitive predictive performance while reducing the need to centralize patient data within the simulated federated learning environment.

D. Comparative Analysis of Centralized and Federated Learning

Fig. 4 compares the classification accuracy of the machine learning models under centralized and simulated federated learning settings. As shown in Fig. 4, the federated framework

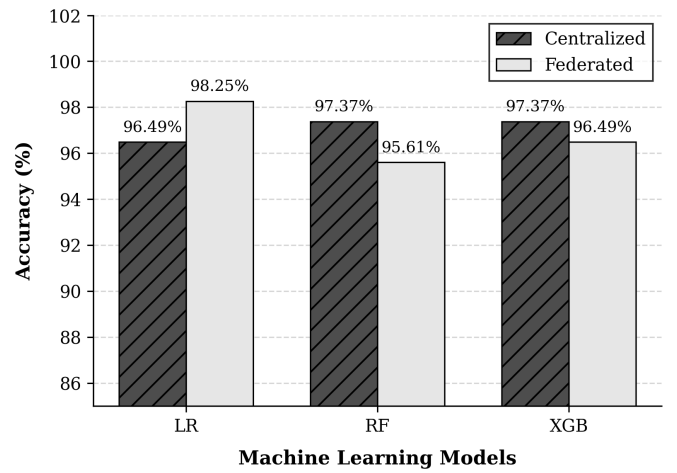


Fig. 4. Comparison of classification accuracy between centralized and federated machine learning models.

achieved performance comparable to the centralized approach

TABLE VIII
FIVE-FOLD CROSS-VALIDATION PERFORMANCE OF CENTRALIZED MODELS.

Model	Acc.	Prec.	Rec.	F1	AUC
LR	97.37 $\pm$ 1.66	98.63 $\pm$ 1.82	94.36 $\pm$ 5.25	96.33 $\pm$ 2.40	0.995 $\pm$ 0.005
RF	95.43 $\pm$ 1.28	94.63 $\pm$ 3.90	93.41 $\pm$ 5.43	93.81 $\pm$ 1.86	0.989 $\pm$ 0.008
XGB	95.26 $\pm$ 0.89	94.22 $\pm$ 4.12	93.41 $\pm$ 4.99	93.60 $\pm$ 1.27	0.994 $\pm$ 0.004

TABLE IX
PERFORMANCE OF FEDERATED GLOBAL MODELS.

Model	Acc.	Prec.	Rec.	F1	AUC
LR	98.25	100.00	95.24	97.56	0.998
RF	95.61	97.44	90.48	93.83	0.995
XGB	96.49	100.00	90.48	95.00	0.998

across all three classifiers. Logistic Regression achieved a higher accuracy in the federated setting (98.25%) than in the centralized setting (96.49%), whereas the Random Forest and XGBoost models also maintained competitive predictive performance using prediction-level probability averaging. The improved predictive capability of Logistic Regression may be attributed to the parameter averaging strategy applied across four nearly balanced client datasets, which can reduce the influence of client-specific variations in this simulated experimental setting. In contrast, the slight decrease observed for Random Forest and XGBoost is consistent with their prediction-level probability averaging strategy, which differs from conventional parameter aggregation. Overall, the results indicate that the proposed framework maintains competitive predictive performance while reducing the need to centralize patient data. However, these observations are based on a simulated federated environment with a single communication round and should not be interpreted as evidence of superior performance in all federated learning scenarios.

E. Performance of the Proposed Soft Voting Ensemble

A Federated Soft Voting Ensemble (FSVE) was constructed by combining the prediction probabilities of Logistic Regression, Random Forest, and XGBoost using equal-weight soft voting. The performance results are summarized in Table 10, while the corresponding ROC curves are presented in Fig. 5. To further assess diagnostic performance, sensitivity and specificity were calculated from the confusion matrices of the best individual federated model and the proposed FSVE, as shown in Table 11. The proposed Federated Soft Voting

TABLE X
PERFORMANCE OF THE PROPOSED FEDERATED SOFT VOTING ENSEMBLE.

Model	Acc.	Prec.	Rec.	F1	ROC-AUC
FSVE	97.37	100.00	92.86	96.30	0.998

Ensemble achieved an accuracy of 97.37%, with 100.00% precision, 92.86% recall, 96.30% F1-score, and a ROC-AUC of 0.998. Although the ensemble did not exceed the accuracy

TABLE XI
DIAGNOSTIC METRICS FOR SELECTED FEDERATED MODELS.

Model	Recall/Sensitivity	Specificity
Federated LR	95.24	100.00
FSVE	92.86	100.00

Receiver Operating Characteristic (ROC) Curves

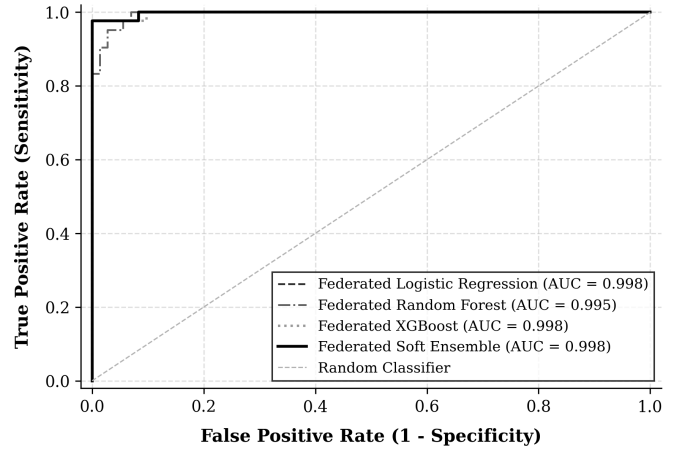


Fig. 5. ROC curves of the federated machine learning models and the proposed FSVE.

of the best-performing individual federated model, Logistic Regression (98.25%), it achieved perfect precision and specificity on the held-out global test set. The ensemble combines the prediction probabilities of the three federated classifiers through equal-weight soft voting, providing a complementary prediction strategy within the simulated federated learning framework.

F. Explainability Analysis

Although the proposed federated learning framework achieved high predictive performance, understanding the factors influencing model predictions is important for improving transparency and interpretability. To support model explainability, SHAP and LIME were applied to the aggregated federated Logistic Regression model. SHAP provides global insights into feature importance by quantifying the contribution of individual features, whereas LIME explains individual predictions by approximating the local decision boundary around a selected instance. Together, these techniques provide complementary global and local explanations of model behavior.

1) *Global Feature Interpretation using SHAP*: Fig. 6 presents the SHAP summary plot illustrating the global importance of diagnostic features for the aggregated federated Logistic Regression model. Features with larger absolute SHAP values have a greater influence on the model's predictions. As shown in Fig. 6, features related to concave points, radius,

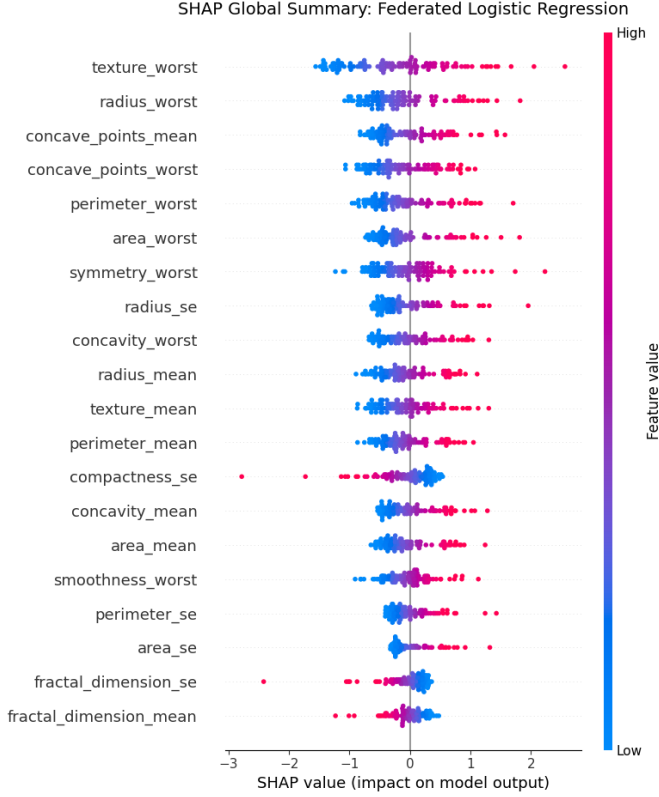


Fig. 6. SHAP summary plot illustrating the global feature importance of the aggregated federated Logistic Regression model.

and perimeter exhibit the highest contribution to distinguishing malignant from benign breast tumors. These observations are consistent with the learned decision patterns of the representative model and demonstrate that the explainability analysis identifies influential diagnostic features used during prediction. Such explanations improve the transparency of the proposed framework and support the interpretation of model predictions.

2) *Local Prediction Interpretation using LIME*: Fig. 7 illustrates the LIME explanation for a representative breast cancer prediction generated by the aggregated federated Logistic Regression model. While SHAP provides global insights into feature importance across the dataset, LIME explains the contribution of individual features to a specific prediction by approximating the model's local decision boundary.

The visualization highlights both positive and negative feature contributions that influence the classification outcome, providing a transparent explanation of the model's prediction for the selected instance. This instance-level explanation complements the global feature importance obtained through SHAP, enabling a more comprehensive interpretation of model

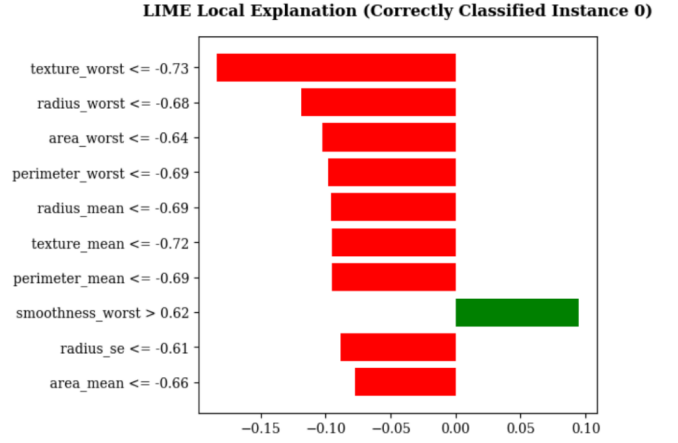


Fig. 7. LIME explanation illustrating the local interpretation of a representative breast cancer prediction.

behavior. Together, SHAP and LIME provide complementary global and local explanations, demonstrating that the proposed framework offers both high predictive performance and improved model transparency.

G. Overall Discussion

Overall, the proposed explainable federated learning framework demonstrates that collaborative model training in a simulated federated environment can achieve predictive performance comparable to centralized learning while reducing the need to centralize patient data. The experimental results show that the federated Logistic Regression model achieved the highest classification performance among the evaluated federated models, whereas the Random Forest and XGBoost models also maintained competitive predictive performance using a prediction-level probability averaging. The proposed Federated Soft Voting Ensemble provided a complementary prediction strategy by combining the outputs of the three base classifiers through equal-weight soft voting. While its accuracy (97.37%) was lower than that of federated Logistic Regression (98.25%), it achieved 100.00% precision and 100.00% specificity on the held-out global test set, while maintaining a ROC-AUC of 0.998. In addition, the integration of SHAP and LIME enhanced the transparency of the framework by providing complementary global and local explanations of model predictions. Although the proposed framework was evaluated using simulated federated clients, a public benchmark dataset, and a single communication round, the results demonstrate the potential of lightweight, explainable, and privacy-aware machine learning approaches for AI-assisted healthcare decision-support and motivate future evaluation in more realistic federated healthcare environments.

V. CONCLUSION AND FUTURE WORK

In this paper, we proposed an explainable federated learning framework for breast cancer diagnosis using the Breast Cancer

Wisconsin (Diagnostic) (WDBC) dataset. The proposed framework combines Federated Learning with Logistic Regression, Random Forest, XGBoost, and a Federated Soft Voting Ensemble to support distributed model training while reducing the need to centralize patient data. Experimental results demonstrated that the simulated federated framework achieved competitive predictive performance, with federated Logistic Regression achieving the highest accuracy of 98.25%. The Federated Soft Voting Ensemble achieved 97.37% accuracy, 100.00% precision, 92.86% recall, 96.30% F1-score, and a ROC-AUC of 0.998. Furthermore, the integration of SHAP and LIME enhanced model transparency by providing complementary global and local explanations of model predictions. Although the framework was evaluated using simulated federated clients on a benchmark dataset with a single communication round, the results demonstrate the potential of lightweight, explainable, and privacy-aware machine learning approaches for AI-assisted healthcare decision support. Future work will focus on evaluating the proposed framework in more realistic federated healthcare environments, including non-IID client distributions, multiple communication rounds, differential privacy mechanisms, and larger real-world healthcare datasets.

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